

JOIcode Side-by-Side Report · gpt41_mini

Side-by-side GT vs Generated JOIcode report for gpt41_mini. This report is rendered from row-level benchmark outputs and is print-friendly for PDF export.

ROWS 19	AVG DET 67.5820	DET PASS 0.0%	GT EXACT 0.0%	AVG PROMPT TOKENS 17814.4
AVG COMPLETION TOKENS 67.8	AVG LLM LATENCY 2.825s	P50 LLM LATENCY 2.218s	PEAK VRAM 0.000 GB	OOM COUNT 0

RUN METADATA

Results Dir	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_by_category_manual_20260506/retrieval_gpt41_mini_cat3
Output Dir	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_by_category_manual_20260506/retrieval_gpt41_mini_cat3/side_by_side_report_gpt41_mini_failures_only
Suite	paper_with_cloud_ref
Genome	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/best_genome.json
Created At	2026-05-06T16:34:35.686824
Categories	3
Limit/Category	-
Rows	61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90

TOP FAILURE REASONS

gt_mismatch	19
dataflow	17
gt_receiver_coverage	7
gt_service_coverage	7
semantic	6
numeric_grounding	5
extraneous	4
precondition	3

CONTEXT + C

service_retrieval
connected_device
Category 3

MODEL RUNTIME SNAPSHOT

MODEL GPT-4.1-mini	RESOLVED MODEL -	WORKER PYTHON -	SERVICE LIST service_list_ver2.0.1.json	CANDIDATE K 1
REPAIR ATTEMPTS 0	DET PROFILE openai			

ROW DETAILS

Row 62 · Category 3

Stop charging if the voltage is 220V or higher.

충전 전압이 220볼트 이상이면, 충전을 멈춰줘.

GT name: StopChargingIfVoltageIs220VOrHigher				
Service list: service_list_ver2.0.1.json				
Output name: StopChargingIfVoltageHigh				
Retrieval cats: Charger, Plug, CarbonDioxideSensor, PresenceVitalSensor, WindowCovering, AudioRecorder, Pump, Safe, ArmRobot, TapDialSwitch				
DET 61.2484	PASS No	GT EXACT No	SIMILARITY 0.8097	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 0.2000	NUMERIC 0.0000	ENUM 1.0000	PROMPT TOKENS 20359
COMPLETION TOKENS 60	TOTAL TOKENS 20419	LLM LATENCY 2.079s	PIPELINE 3.219s	TOK/S 9822.0213
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

PROMPT CONTEXT SNAPSHOT

Retrieval premapping was used for this row. Requested top-10 and injected 10 category group(s) into the prompt.

CONTEXT MODE retrieval premapping	SNIPPET SOURCE service_retrieval_fallback	RETRIEVAL used	CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10	INJECTED CATEGORIES 10
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Charger

RETRIEVED · RANK 1

4 services · 4 values · 0 functions

VALUES

Charger_ChargingState

raw ChargingState · arg - · ret ENUM · The current status of battery charging

Charger_Current

raw Current · arg - · ret DOUBLE · The current status of battery charging

Charger_Power

raw Power · arg - · ret DOUBLE · The current status of battery charging

Charger_Voltage

raw Voltage · arg - · ret DOUBLE · The current status of battery charging

FUNCTIONS

None

Plug

RETRIEVED · RANK 2

3 services · 3 values · 0 functions

VALUES

Plug_Current

raw Current · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

Plug_Power

raw Power · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

Plug_Voltage

raw Voltage · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

FUNCTIONS

None

CarbonDioxideSensor

RETRIEVED · RANK 3

1 services · 1 values · 0 functions

VALUES

CarbonDioxideSensor_CarbonDioxide

raw CarbonDioxide · arg - · ret DOUBLE · Measure carbon dioxide levels

FUNCTIONS

None

PresenceVitalSensor

RETRIEVED · RANK 4

7 services · 7 values · 0 functions

VALUES

PresenceVitalSensor_Awakeness

raw Awakeness · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Distance

raw Distance · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_DwellTime

raw DwellTime · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_HeartRate

raw HeartRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_MovementIndex

raw MovementIndex · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Presence

raw Presence · arg - · ret BOOL · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_RespiratoryRate

raw RespiratoryRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

FUNCTIONS

None

WindowCovering

RETRIEVED · RANK 5

6 services · 2 values · 4 functions

VALUES

WindowCovering_CurrentPosition

raw CurrentPosition · arg - · ret INTEGER · Controls a window covering to open or close it

WindowCovering_WindowCoveringType

raw WindowCoveringType · arg - · ret ENUM · Controls a window covering to open or close it

FUNCTIONS

WindowCovering_DownOrClose

raw DownOrClose · arg - · ret VOID · Controls a window covering to open or close it

WindowCovering_SetLevel

raw SetLevel · arg INTEGER · ret INTEGER · Controls a window covering to open or close it

WindowCovering_Stop

raw Stop · arg - · ret VOID · Controls a window covering to open or close it

WindowCovering_UpOrOpen

raw UpOrOpen · arg - · ret VOID · Controls a window covering to open or close it

AudioRecorder

RETRIEVED · RANK 6

5 services · 2 values · 3 functions

VALUES

AudioRecorder_AudioFile

raw AudioFile · arg - · ret BINARY · Record audio

AudioRecorder_RecordStatus

raw RecordStatus · arg - · ret ENUM · Record audio

FUNCTIONS

AudioRecorder_RecordStart

raw RecordStart · arg - · ret VOID · Record audio

AudioRecorder_RecordStop

raw RecordStop · arg BINARY · ret VOID · Record audio

AudioRecorder_RecordWithDuration

raw RecordWithDuration · arg STRING | DOUBLE · ret BINARY · Record audio

Pump

RETRIEVED · RANK 7

2 services · 1 values · 1 functions

VALUES

Pump_PumpMode
raw PumpMode · arg - · ret ENUM · Allows for the control of a pump device

FUNCTIONS

Pump_SetPumpMode
raw SetPumpMode · arg ENUM · ret VOID · Allows for the control of a pump device

Safe

3 services · 1 values · 2 functions

RETRIEVED · RANK 8

VALUES

Safe_SafeState
raw SafeState · arg - · ret ENUM · Allows for the control of the Safe

FUNCTIONS

Safe_Lock
raw Lock · arg - · ret VOID · Allows for the control of the Safe

Safe_Unlock
raw Unlock · arg - · ret VOID · Allows for the control of the Safe

ArmRobot

5 services · 2 values · 3 functions

RETRIEVED · RANK 9

VALUES

ArmRobot_ArmRobotType
raw ArmRobotType · arg - · ret ENUM · Allows for the control of the arm robot

ArmRobot_CurrentPosition
raw CurrentPosition · arg - · ret STRING · Allows for the control of the arm robot

FUNCTIONS

ArmRobot_Hello
raw Hello · arg - · ret VOID · Allows for the control of the arm robot

ArmRobot_SendCommand
raw SendCommand · arg ENUM · ret VOID · Allows for the control of the arm robot

ArmRobot_SetPosition
raw SetPosition · arg ENUM · ret VOID · Allows for the control of the arm robot

TapDialSwitch

6 services · 6 values · 0 functions

RETRIEVED · RANK 10

VALUES

TapDialSwitch_Button1
raw Button1 · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_Button2
raw Button2 · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_Button3

raw Button3 · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_Button4

raw Button4 · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_Rotation

raw Rotation · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_RotationSteps

raw RotationSteps · arg - · ret INTEGER · A tap dial switch device with multiple buttons and a rotary dial

FUNCTIONS

None

GT JOICODE

cron="" · period=0

```
if ((#Charger).charger_voltage >= 220000) {  
  
    (#Charger).switch_off()  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#Charger).charger_voltage >= 220.0) {  
    (#Charger).safe_lock()  
}
```

FAILURE REASONS

- dataflow
- extraneous
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- numeric_grounding
- semantic

LINE DIFF

GT	Generated
if ((#Charger).charger_voltage >= 220000) {	if ((#Charger).charger_voltage >= 220.0) {
	(#Charger).safe_lock()
(#Charger).switch_off()	}
}	

Row 63 · Category 3

If the cloud service is activated, upload test.png file to the cloud.

클라우드 서비스가 활성화되어있으면, test.png 파일을 클라우드에 업로드해줘.

GT name: IfCloudServiceIsActivatedUploadTestPngFile				
Service list: service_list_ver2.0.1.json				
Output name: UploadFileWhenCloudServiceActivated				
Retrieval cats: CloudServiceProvider, WeatherProvider, LevelControl, MenuProvider, EmailProvider, AirPurifier, Charger, Button, Pump, Plug				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.8313	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 1.0000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 20975
COMPLETION TOKENS 68	TOTAL TOKENS 21043	LLM LATENCY 2.062s	PIPELINE 3.127s	TOK/S 10207.6158
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

PROMPT CONTEXT SNAPSHOT

Retrieval premapping was used for this row. Requested top-10 and injected 10 category group(s) into the prompt.

CONTEXT MODE retrieval premapping	SNIPPET SOURCE service_retrieval_fallback	RETRIEVAL used	CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10	INJECTED CATEGORIES 10
--------------------------------------	--	-------------------	------------------------	-----------------------	---------------------------

CloudServiceProvider

14 services · 5 values · 9 functions

VALUES

CloudServiceProvider_ChatSession

raw ChatSession · arg - · ret STRING · Provides cloud service functionalities

CloudServiceProvider_GeneratedImage

raw GeneratedImage · arg - · ret BINARY · Provides cloud service functionalities

CloudServiceProvider_ImageExplanation

raw ImageExplanation · arg - · ret STRING · Provides cloud service functionalities

CloudServiceProvider_LLMModels

RETRIEVED · RANK 1

raw LLMModels · arg - · ret LIST · Provides cloud service functionalities

CloudServiceProvider_UploadedFile

raw UploadedFile · arg - · ret BINARY · Provides cloud service functionalities

FUNCTIONS

CloudServiceProvider_ChatWithAI

raw ChatWithAI · arg STRING · ret STRING · Provides cloud service functionalities

CloudServiceProvider_ExplainImage

raw ExplainImage · arg BINARY · ret STRING · Provides cloud service functionalities

CloudServiceProvider_GenerateImage

raw GenerateImage · arg STRING · ret BINARY · Provides cloud service functionalities

CloudServiceProvider_IsAvailable

raw IsAvailable · arg BOOL · ret BOOL · Provides cloud service functionalities

CloudServiceProvider_SaveToFile

raw SaveToFile · arg BINARY | STRING · ret STRING · Provides cloud service functionalities

CloudServiceProvider_SpeechToText

raw SpeechToText · arg STRING · ret STRING · Provides cloud service functionalities

CloudServiceProvider_TextToSpeech

raw TextToSpeech · arg STRING · ret BINARY · Provides cloud service functionalities

CloudServiceProvider_UploadFile

raw UploadFile · arg STRING · ret BINARY · Provides cloud service functionalities

CloudServiceProvider_UploadToCloudStorage

raw UploadToCloudStorage · arg STRING · ret STRING · Provides cloud service functionalities

WeatherProvider

RETRIEVED · RANK 2

7 services · 6 values · 1 functions

VALUES

WeatherProvider_HumidityWeather

raw HumidityWeather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_Pm10Weather

raw Pm10Weather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_Pm25Weather

raw Pm25Weather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_PressureWeather

raw PressureWeather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_TemperatureWeather

raw TemperatureWeather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_Weather

raw Weather · arg - · ret ENUM · Provides weather information

FUNCTIONS

WeatherProvider_GetWeatherInfo
raw GetWeatherInfo · arg DOUBLE | DOUBLE · ret STRING · Provides weather information

LevelControl

RETRIEVED · RANK 3

4 services · 3 values · 1 functions

VALUES

LevelControl_CurrentLevel
raw CurrentLevel · arg - · ret DOUBLE · Allows for the control of the level of a device like a light or a dimmer switch

LevelControl_MaxLevel
raw MaxLevel · arg - · ret DOUBLE · Allows for the control of the level of a device like a light or a dimmer switch

LevelControl_MinLevel
raw MinLevel · arg - · ret DOUBLE · Allows for the control of the level of a device like a light or a dimmer switch

FUNCTIONS

LevelControl_MoveToLevel
raw MoveToLevel · arg DOUBLE | DOUBLE · ret VOID · Allows for the control of the level of a device like a light or a dimmer switch

MenuProvider

RETRIEVED · RANK 4

4 services · 3 values · 1 functions

VALUES

MenuProvider_Menu
raw Menu · arg - · ret STRING · Provides menu information services

MenuProvider_TodayMenu
raw TodayMenu · arg - · ret STRING · Provides menu information services

MenuProvider_TodayPlace
raw TodayPlace · arg - · ret STRING · Provides menu information services

FUNCTIONS

MenuProvider_GetMenu
raw GetMenu · arg STRING · ret STRING · Provides menu information services

EmailProvider

RETRIEVED · RANK 5

2 services · 0 values · 2 functions

VALUES

None

FUNCTIONS

EmailProvider_SendMail
raw SendMail · arg STRING | STRING | STRING · ret VOID · Provides email service

EmailProvider_SendMailWithFile
raw SendMailWithFile · arg STRING | STRING | STRING | STRING · ret VOID · Provides email service

AirPurifier

RETRIEVED · RANK 6

2 services · 1 values · 1 functions

VALUES

AirPurifier_AirPurifierMode

raw AirPurifierMode · arg - · ret ENUM · Controls air purifier mode

FUNCTIONS

AirPurifier_SetAirPurifierMode

raw SetAirPurifierMode · arg ENUM · ret VOID · Controls air purifier mode

Charger

RETRIEVED · RANK 7

4 services · 4 values · 0 functions

VALUES

Charger_ChargingState

raw ChargingState · arg - · ret ENUM · The current status of battery charging

Charger_Current

raw Current · arg - · ret DOUBLE · The current status of battery charging

Charger_Power

raw Power · arg - · ret DOUBLE · The current status of battery charging

Charger_Voltage

raw Voltage · arg - · ret DOUBLE · The current status of battery charging

FUNCTIONS

None

Button

RETRIEVED · RANK 8

1 services · 1 values · 0 functions

VALUES

Button_Button

raw Button · arg - · ret ENUM · A device with one or more buttons

FUNCTIONS

None

Pump

RETRIEVED · RANK 9

2 services · 1 values · 1 functions

VALUES

Pump_PumpMode

raw PumpMode · arg - · ret ENUM · Allows for the control of a pump device

FUNCTIONS

Pump_SetPumpMode

raw SetPumpMode · arg ENUM · ret VOID · Allows for the control of a pump device

Plug

RETRIEVED · RANK 10

3 services · 3 values · 0 functions

VALUES

Plug_Current

raw Current · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

Plug_Power

raw Power · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

Plug_Voltage

raw Voltage · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

FUNCTIONS

None

GT JOICODE

cron="" · period=0

```
if ((#CloudServiceProvider).cloudserviceprovider_isavailable == true) {  
    (#CloudServiceProvider).cloudserviceprovider_uploadfile("test.png")  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#CloudServiceProvider).cloudserviceprovider_chatsession != "") {  
    (#CloudServiceProvider).cloudserviceprovider_uploadfile("test.png")  
}
```

FAILURE REASONS

- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage

LINE DIFF

GT	Generated
if ((#CloudServiceProvider).cloudserviceprovider_isavailable == true) {	if ((#CloudServiceProvider).cloudserviceprovider_chatsession != "") {
	(#CloudServiceProvider).cloudserviceprovider_uploadfile("test.png")
(#CloudServiceProvider).cloudserviceprovider_uploadfile("test.png")	}
}	

Row 64 · Category 3

If the oven is in grill mode, add 4 more minutes.

오븐이 그릴 모드이면, 시간을 4분 더 추가해줘

GT name: IfOvenIsInGrillModeAdd4More				
Service list: service_list_ver2.0.1.json				
Output name: AddTimelfOvenInGrillMode				
Retrieval cats: Oven, RiceCooker, MenuProvider, Dishwasher, Siren, Clock, DimmerSwitch, AirConditioner, Plug, SmokeDetector				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9930	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 0.0000	ENUM 1.0000	PROMPT TOKENS 20511
COMPLETION TOKENS 67	TOTAL TOKENS 20578	LLM LATENCY 2.902s	PIPELINE 3.847s	TOK/S 7089.9945
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

PROMPT CONTEXT SNAPSHOT

Retrieval premapping was used for this row. Requested top-10 and injected 10 category group(s) into the prompt.

CONTEXT MODE retrieval premapping	SNIPPET SOURCE service_retrieval_fallback	RETRIEVAL used	CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10	INJECTED CATEGORIES 10
--------------------------------------	--	-------------------	------------------------	-----------------------	---------------------------

Oven

4 services · 1 values · 3 functions

VALUES

Oven_OvenMode

raw OvenMode · arg - · ret ENUM · Allows for the control of the oven mode

FUNCTIONS

Oven_AddMoreTime

raw AddMoreTime · arg DOUBLE · ret VOID · Allows for the control of the oven mode

Oven_SetCookingParameters

raw SetCookingParameters · arg ENUM | DOUBLE · ret VOID · Allows for the control of the oven mode

RETRIEVED · RANK 1

Oven_SetOvenMode

raw SetOvenMode · arg ENUM · ret VOID · Allows for the control of the oven mode

RiceCooker

RETRIEVED · RANK 2

4 services · 1 values · 3 functions

VALUES

RiceCooker_RiceCookerMode

raw RiceCookerMode · arg - · ret ENUM · Allows for the control of the Rice Cooker

FUNCTIONS

RiceCooker_AddMoreTime

raw AddMoreTime · arg DOUBLE · ret VOID · Allows for the control of the Rice Cooker

RiceCooker_SetCookingParameters

raw SetCookingParameters · arg ENUM | DOUBLE · ret VOID · Allows for the control of the Rice Cooker

RiceCooker_SetRiceCookerMode

raw SetRiceCookerMode · arg ENUM · ret VOID · Allows for the control of the Rice Cooker

MenuProvider

RETRIEVED · RANK 3

4 services · 3 values · 1 functions

VALUES

MenuProvider_Menu

raw Menu · arg - · ret STRING · Provides menu information services

MenuProvider_TodayMenu

raw TodayMenu · arg - · ret STRING · Provides menu information services

MenuProvider_TodayPlace

raw TodayPlace · arg - · ret STRING · Provides menu information services

FUNCTIONS

MenuProvider_GetMenu

raw GetMenu · arg STRING · ret STRING · Provides menu information services

Dishwasher

RETRIEVED · RANK 4

2 services · 1 values · 1 functions

VALUES

Dishwasher_DishwasherMode

raw DishwasherMode · arg - · ret ENUM · Allows for the control of the dishwasher mode

FUNCTIONS

Dishwasher_SetDishwasherMode

raw SetDishwasherMode · arg ENUM · ret VOID · Allows for the control of the dishwasher mode

Siren

RETRIEVED · RANK 5

2 services · 1 values · 1 functions

VALUES

Siren_SirenMode

raw SirenMode · arg - · ret ENUM · Allows for the control of the Siren

FUNCTIONS

Siren_SetSirenMode

raw SetSirenMode · arg ENUM · ret VOID · Allows for the control of the Siren

Clock

13 services · 12 values · 1 functions

RETRIEVED · RANK 6

VALUES

Clock_Date

raw Date · arg - · ret STRING · Provide current date and time

Clock_Datetime

raw Datetime · arg - · ret STRING · Provide current date and time

Clock_Day

raw Day · arg - · ret INTEGER · Provide current date and time

Clock_Hour

raw Hour · arg - · ret INTEGER · Provide current date and time

Clock_IsHoliday

raw IsHoliday · arg - · ret BOOL · Provide current date and time

Clock_Minute

raw Minute · arg - · ret INTEGER · Provide current date and time

Clock_Month

raw Month · arg - · ret INTEGER · Provide current date and time

Clock_Second

raw Second · arg - · ret INTEGER · Provide current date and time

Clock_Time

raw Time · arg - · ret STRING · Provide current date and time

Clock_Timestamp

raw Timestamp · arg - · ret DOUBLE · Provide current date and time

Clock_Weekday

raw Weekday · arg - · ret ENUM · Provide current date and time

Clock_Year

raw Year · arg - · ret INTEGER · Provide current date and time

FUNCTIONS

Clock_Delay

raw Delay · arg INTEGER | INTEGER | INTEGER · ret VOID · Provide current date and time

DimmerSwitch

4 services · 4 values · 0 functions

RETRIEVED · RANK 7

VALUES

DimmerSwitch_Button1

raw Button1 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

DimmerSwitch_Button2

raw Button2 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

DimmerSwitch_Button3

raw Button3 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

DimmerSwitch_Button4

raw Button4 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

FUNCTIONS

None

AirConditioner

4 services · 2 values · 2 functions

RETRIEVED · RANK 8

VALUES

AirConditioner_AirConditionerMode

raw AirConditionerMode · arg - · ret ENUM · Controls air conditioner mode and temperature settings

AirConditioner_TargetTemperature

raw TargetTemperature · arg - · ret DOUBLE · Controls air conditioner mode and temperature settings

FUNCTIONS

AirConditioner_SetAirConditionerMode

raw SetAirConditionerMode · arg ENUM · ret VOID · Controls air conditioner mode and temperature settings

AirConditioner_SetTargetTemperature

raw SetTargetTemperature · arg DOUBLE · ret VOID · Controls air conditioner mode and temperature settings

Plug

3 services · 3 values · 0 functions

RETRIEVED · RANK 9

VALUES

Plug_Current

raw Current · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

Plug_Power

raw Power · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

Plug_Voltage

raw Voltage · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

FUNCTIONS

None

SmokeDetector

1 services · 1 values · 0 functions

RETRIEVED · RANK 10

VALUES

SmokeDetector_Smoke
raw Smoke · arg - · ret BOOL · A Device that detects smoke

FUNCTIONS
None

GT JOICODE

```
cron="" · period=0

if ((#0ven).oven_ovenmode == "grill") {

    (#0ven).oven_addmoretime(240)

}
```

GENERATED JOICODE

```
cron="" · period=0

if ((#0ven).oven_ovenmode == "grill") {
    (#0ven).oven_addmoretime(4)
}
```

FAILURE REASONS

- dataflow
- gt_mismatch
- numeric_grounding

LINE DIFF

GT	Generated
if ((#0ven).oven_ovenmode == "grill") {	if ((#0ven).oven_ovenmode == "grill") {
	(#0ven).oven_addmoretime(4)
(#0ven).oven_addmoretime(240)	}
}	

Row 74 · Category 3

If the button is in the pushed state, raise the blind.

버튼이 눌러진 상태면, 블라인드를 올려줘.

GT name: IfButtonIsInPushedStateRaise				
Service list: service_list_ver2.0.1.json				
Output name: RaiseBlindWhenButtonPushed				
Retrieval cats: Button, DimmerSwitch, TapDialSwitch, PresenceVitalSensor, ContactSensor, Safe, Plug, PresenceSensor, Charger, ArmRobot				
DET 59.2874	PASS No	GT EXACT No	SIMILARITY 0.8175	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 0.0000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 19813
COMPLETION TOKENS 54	TOTAL TOKENS 19867	LLM LATENCY 2.429s	PIPELINE 3.447s	TOK/S 8177.7394
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

PROMPT CONTEXT SNAPSHOT

Retrieval premapping was used for this row. Requested top-10 and injected 10 category group(s) into the prompt.

CONTEXT MODE retrieval premapping	SNIPPET SOURCE service_retrieval_fallback	RETRIEVAL used	CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10	INJECTED CATEGORIES 10
--------------------------------------	--	-------------------	------------------------	-----------------------	---------------------------

Button

1 services · 1 values · 0 functions

VALUES

Button_Button
raw Button · arg - · ret ENUM · A device with one or more buttons

FUNCTIONS

None

RETRIEVED · RANK 1

DimmerSwitch

4 services · 4 values · 0 functions

RETRIEVED · RANK 2

VALUES

DimmerSwitch_Button1

raw Button1 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

DimmerSwitch_Button2

raw Button2 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

DimmerSwitch_Button3

raw Button3 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

DimmerSwitch_Button4

raw Button4 · arg - · ret ENUM · A dimmer switch device with multiple buttons (typically 4 buttons)

FUNCTIONS

None

TapDialSwitch

RETRIEVED · RANK 3

6 services · 6 values · 0 functions

VALUES

TapDialSwitch_Button1

raw Button1 · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_Button2

raw Button2 · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_Button3

raw Button3 · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_Button4

raw Button4 · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_Rotation

raw Rotation · arg - · ret ENUM · A tap dial switch device with multiple buttons and a rotary dial

TapDialSwitch_RotationSteps

raw RotationSteps · arg - · ret INTEGER · A tap dial switch device with multiple buttons and a rotary dial

FUNCTIONS

None

PresenceVitalSensor

RETRIEVED · RANK 4

7 services · 7 values · 0 functions

VALUES

PresenceVitalSensor_Awakeness

raw Awakeness · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Distance

raw Distance · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_DwellTime

raw DwellTime · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_HeartRate

raw HeartRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_MovementIndex

raw MovementIndex · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_Presence

raw Presence · arg - · ret BOOL · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

PresenceVitalSensor_RespiratoryRate

raw RespiratoryRate · arg - · ret DOUBLE · Presence and vital signs sensor with heart rate, respiratory rate, movement detection

FUNCTIONS

None

ContactSensor

RETRIEVED · RANK 5

1 services · 1 values · 0 functions

VALUES

ContactSensor_Contact

raw Contact · arg - · ret BOOL · Allows reading the value of a contact sensor device

FUNCTIONS

None

Safe

RETRIEVED · RANK 6

3 services · 1 values · 2 functions

VALUES

Safe_SafeState

raw SafeState · arg - · ret ENUM · Allows for the control of the Safe

FUNCTIONS

Safe_Lock

raw Lock · arg - · ret VOID · Allows for the control of the Safe

Safe_Unlock

raw Unlock · arg - · ret VOID · Allows for the control of the Safe

Plug

RETRIEVED · RANK 7

3 services · 3 values · 0 functions

VALUES

Plug_Current

raw Current · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

Plug_Power

raw Power · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

Plug_Voltage

raw Voltage · arg - · ret DOUBLE · Allows for monitoring power consumption of a plug device

FUNCTIONS

None

PresenceSensor

RETRIEVED · RANK 8

1 services · 1 values · 0 functions

VALUES

PresenceSensor_Presence

raw Presence · arg - · ret BOOL · The ability to see the current status of a presence sensor device

FUNCTIONS

None

Charger

RETRIEVED · RANK 9

4 services · 4 values · 0 functions

VALUES

Charger_ChargingState

raw ChargingState · arg - · ret ENUM · The current status of battery charging

Charger_Current

raw Current · arg - · ret DOUBLE · The current status of battery charging

Charger_Power

raw Power · arg - · ret DOUBLE · The current status of battery charging

Charger_Voltage

raw Voltage · arg - · ret DOUBLE · The current status of battery charging

FUNCTIONS

None

ArmRobot

RETRIEVED · RANK 10

5 services · 2 values · 3 functions

VALUES

ArmRobot_ArmRobotType

raw ArmRobotType · arg - · ret ENUM · Allows for the control of the arm robot

ArmRobot_CurrentPosition

raw CurrentPosition · arg - · ret STRING · Allows for the control of the arm robot

FUNCTIONS

ArmRobot_Hello

raw Hello · arg - · ret VOID · Allows for the control of the arm robot

ArmRobot_SendCommand

raw SendCommand · arg ENUM · ret VOID · Allows for the control of the arm robot

ArmRobot_SetPosition

raw SetPosition · arg ENUM · ret VOID · Allows for the control of the arm robot

GT JOICODE

cron="" · period=0

GENERATED JOICODE

cron="" · period=0

```
if ((#Button).button_button == "pushed") {  
  
    (#Blind).windowcovering_uporopen()  
  
}
```

```
if ((#Button).button_button == "pushed") {  
    (#Blind).blind_open()  
}
```

FAILURE REASONS

- arg_type
- dataflow
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- semantic
- service_match
- unknown_service:blind_open

LINE DIFF

GT	Generated
if ((#Button).button_button == "pushed") {	if ((#Button).button_button == "pushed") {
	(#Blind).blind_open()
(#Blind).windowcovering_uporopen()	}
}	

Row 76 · Category 3

If face recognition at the entrance is off, start it.

입구의 얼굴 인식이 꺼져있으면, 얼굴 인식을 시작해줘

GT name: IfFaceRecognitionAtEntranceIsOffStartIt				
Service list: service_list_ver2.0.1.json				
Output name: StartFaceRecognitionIfOff				
Retrieval cats: not used (connected_devices scope)				
DET 63.5886	PASS No	GT EXACT No	SIMILARITY 0.7177	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 16949
COMPLETION TOKENS 59	TOTAL TOKENS 17008	LLM LATENCY 1.786s	PIPELINE 1.794s	TOK/S 9525.6231
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 2	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
---	--	--	------------------------	----------------------	--------------------------

raw DeleteFace · arg STRING · ret BOOL · Controls face recognition features

FaceRecognizer_End

raw End · arg - · ret BOOL · Controls face recognition features

FaceRecognizer_Start

raw Start · arg - · ret BOOL · Controls face recognition features

Switch

4 services · 1 values · 3 functions

CONNECTED

VALUES

Switch_Switch

raw Switch · arg - · ret BOOL · Allows for the control of a Switch device

FUNCTIONS

Switch_Off

raw Off · arg - · ret VOID · Allows for the control of a Switch device

Switch_On

raw On · arg - · ret VOID · Allows for the control of a Switch device

Switch_Toggle

raw Toggle · arg - · ret BOOL · Allows for the control of a Switch device

GT JOICODE

cron="" · period=0

```
if ((#Entrance #FaceRecognizer).switch_switch == false) {  
    (#Entrance #FaceRecognizer).facerecognizer_start()  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#FaceRecognizer).facerecognizer_recognizedresult == "off") {  
    (#FaceRecognizer).facerecognizer_start()  
}
```

FAILURE REASONS

dataflow

gt_mismatch

gt_receiver_coverage

gt_service_coverage

precondition

semantic

LINE DIFF

GT	Generated
if ((#Entrance #FaceRecognizer).switch_switch == false) {	if ((#FaceRecognizer).facerecognizer_recognizedresult == "off") {
	(#FaceRecognizer).facerecognizer_start()
(#Entrance #FaceRecognizer).facerecognizer_start()	}
}	

Row 77 · Category 3

If the shade is closed to 30% or less, close it to 50%.

쉐이드가 30%이하로 닫혀있으면, 50%만큼 닫아줘.

GT name: IfShadeIsClosedTo30OrLessClose				
Service list: service_list_ver2.0.1.json				
Output name: ShadePositionAdjustment				
Retrieval cats: not used (connected_devices scope)				
DET 61.8872	PASS No	GT EXACT No	SIMILARITY 0.6774	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 16792
COMPLETION TOKENS 79	TOTAL TOKENS 16871	LLM LATENCY 5.598s	PIPELINE 5.605s	TOK/S 3013.8626
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 2	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 1 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 1
---	--	--	------------------------	----------------------	--------------------------

WindowCovering

CONNECTED

6 services · 2 values · 4 functions

VALUES

WindowCovering_CurrentPosition

raw CurrentPosition · arg - · ret INTEGER · Controls a window covering to open or close it

WindowCovering_WindowCoveringType

raw WindowCoveringType · arg - · ret ENUM · Controls a window covering to open or close it

FUNCTIONS

WindowCovering_DownOrClose

raw DownOrClose · arg - · ret VOID · Controls a window covering to open or close it

WindowCovering_SetLevel

raw SetLevel · arg INTEGER · ret INTEGER · Controls a window covering to open or close it

WindowCovering_Stop

raw Stop · arg - · ret VOID · Controls a window covering to open or close it

WindowCovering_UpOrOpen

raw UpOrOpen · arg - · ret VOID · Controls a window covering to open or close it

GT JOICODE

cron="" · period=0

```
if ((#Shade).armrobot_currentposition <= 30) {  
  
    (#Shade).windowcovering_setLevel(50)  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#WindowCovering).windowcovering_windowcoveringtype == "shade" and  
(#WindowCovering).windowcovering_currentposition <= 30) {  
    (#WindowCovering).windowcovering_setLevel(50)  
}
```

FAILURE REASONS

- dataflow
- extraneous
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- semantic

LINE DIFF

GT	Generated
if ((#Shade).armrobot_currentposition <= 30) {	if ((#WindowCovering).windowcovering_windowcoveringtype == "shade" and (#WindowCovering).windowcovering_currentposition <= 30) {
	(#WindowCovering).windowcovering_setLevel(50)
(#Shade).windowcovering_setLevel(50)	}
}	

Row 78 · Category 3

If the sound sensor level in livingroom is over 50, stop the speaker.

거실의 소리 센서로 감지한 수치가 50보다 크면, 스피커 재생을 멈춰줘.

GT name: IfSoundSensorLevelInLivingroomIsOver50				
Service list: service_list_ver2.0.1.json				
Output name: StopSpeakerWhenSoundHigh				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.8800	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 17292
COMPLETION TOKENS 61	TOTAL TOKENS 17353	LLM LATENCY 2.218s	PIPELINE 2.225s	TOK/S 7824.0678
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
---	--	--	------------------------	----------------------	--------------------------

SoundSensor

1 services · 1 values · 0 functions

VALUES

SoundSensor_Sound

raw Sound · arg · ret DOUBLE · Sound sensor device for measuring sound levels

FUNCTIONS

None

Speaker

CONNECTED

11 services · 2 values · 9 functions

VALUES

Speaker_PlaybackState

raw PlaybackState · arg - · ret ENUM · Speaker device for audio playback and media control

Speaker_Volume

raw Volume · arg - · ret INTEGER · Speaker device for audio playback and media control

FUNCTIONS

Speaker_FastForward

raw FastForward · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_Pause

raw Pause · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_Play

raw Play · arg STRING · ret VOID · Speaker device for audio playback and media control

Speaker_Rewind

raw Rewind · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_SetVolume

raw SetVolume · arg INTEGER · ret INTEGER · Speaker device for audio playback and media control

Speaker_Speak

raw Speak · arg STRING · ret VOID · Speaker device for audio playback and media control

Speaker_Stop

raw Stop · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_VolumeDown

raw VolumeDown · arg - · ret INTEGER · Speaker device for audio playback and media control

Speaker_VolumeUp

raw VolumeUp · arg - · ret INTEGER · Speaker device for audio playback and media control

GT JOICODE

cron="" · period=0

```
if ((#LivingRoom #SoundSensor).soundsensor_sound > 50) {  
  
    (#LivingRoom #Speaker).speaker_stop()  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#SoundSensor #Livingroom).soundsensor_sound > 50) {  
    (#Speaker #Livingroom).speaker_stop()  
}
```

FAILURE REASONS

dataflow

gt_mismatch

LINE DIFF

GT	Generated
if ((#LivingRoom #SoundSensor).soundsensor_sound > 50) {	if ((#SoundSensor #Livingroom).soundsensor_sound > 50) {
	(#Speaker #Livingroom).speaker_stop()

GT	Generated
(#LivingRoom #Speaker).speaker_stop()	}
}	

Row 79 · Category 3

If the air conditioner is in cool mode, switch it to auto mode.

거실의 에어컨 모드가 냉방 모드이면, 자동 모드로 바꿔줘.

GT name: IfAirConditionerIsInCoolModeSwitchIt				
Service list: service_list_ver2.0.1.json				
Output name: AirConditioner_CoolToAutoMode				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9612	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 16609
COMPLETION TOKENS 74	TOTAL TOKENS 16683	LLM LATENCY 5.579s	PIPELINE 5.586s	TOK/S 2990.3744
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 2	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 1 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 1
---	--	--	------------------------	----------------------	--------------------------

AirConditioner

CONNECTED

4 services · 2 values · 2 functions

VALUES

AirConditioner_AirConditionerMode

raw AirConditionerMode · arg - · ret ENUM · Controls air conditioner mode and temperature settings

AirConditioner_TargetTemperature

raw TargetTemperature · arg - · ret DOUBLE · Controls air conditioner mode and temperature settings

FUNCTIONS

AirConditioner_SetAirConditionerMode

raw SetAirConditionerMode · arg ENUM · ret VOID · Controls air conditioner mode and temperature settings

AirConditioner_SetTargetTemperature

raw SetTargetTemperature · arg DOUBLE · ret VOID · Controls air conditioner mode and temperature settings

GT JOICODE

cron="" · period=0

```
if ((#LivingRoom #AirConditioner).airconditioner_airconditionermode == "cool") {  
  
    (#LivingRoom #AirConditioner).airconditioner_setairconditionermode("auto")  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#AirConditioner).airconditioner_airconditionermode == "cool") {  
    (#AirConditioner).airconditioner_setairconditionermode("auto")  
}
```

FAILURE REASONS

- dataflow
- gt_mismatch

LINE DIFF

GT	Generated
if ((#LivingRoom #AirConditioner).airconditioner_airconditionermode == "cool") {	if ((#AirConditioner).airconditioner_airconditionermode == "cool") {
	(#AirConditioner).airconditioner_setairconditionermode("auto")
(#LivingRoom #AirConditioner).airconditioner_setairconditionermode("auto")	}
}	

Row 80 · Category 3

If any presence sensor in the house is currently in the detected state, set the siren to emergency mode.

집에 있는 재질 센서중 하나라도 감지 상태이면, 긴급 사이렌을 울려줘.

GT name: IfAnyPresenceSensorInHousesCurrentlyIn				
Service list: service_list_ver2.0.1.json				
Output name: PresenceDetectedTriggerEmergencySiren				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.8209	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 16515
COMPLETION TOKENS 66	TOTAL TOKENS 16581	LLM LATENCY 1.755s	PIPELINE 1.762s	TOK/S 9446.2485
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
---	--	--	------------------------	----------------------	--------------------------

PresenceSensor 1 services · 1 values · 0 functions VALUES PresenceSensor_Presence raw Presence · arg - · ret BOOL · The ability to see the current status of a presence sensor device FUNCTIONS None	CONNECTED
Siren	CONNECTED

2 services · 1 values · 1 functions

VALUES

Siren_SirenMode

raw SirenMode · arg - · ret ENUM · Allows for the control of the Siren

FUNCTIONS

Siren_SetSirenMode

raw SetSirenMode · arg ENUM · ret VOID · Allows for the control of the Siren

GT JOICODE

cron="" · period=0

```
if (all(#House #PresenceSensor).presencesensor_presence ==| true) {  
  
    (#Siren).siren_setsirenmode("emergency")  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if (any(#PresenceSensor).presencesensor_presence == true) {  
    (#Siren).siren_setsirenmode("emergency")  
}
```

FAILURE REASONS

- dataflow
- gt_mismatch

LINE DIFF

GT	Generated
if (all(#House #PresenceSensor).presencesensor_presence == true) {	if (any(#PresenceSensor).presencesensor_presence == true) {
	(#Siren).siren_setsirenmode("emergency")
(#Siren).siren_setsirenmode("emergency")	}
}	

Row 81 · Category 3

If the contact sensor at the entrance is in detected state, sound the emergency siren.

입구의 접촉 센서가 감지 상태이면, 긴급 사이렌을 울려줘.

GT name: IfContactSensorAtEntrancelInDetected				
Service list: service_list_ver2.0.1.json				
Output name: EntranceContactEmergencySiren				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9763	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 16519
COMPLETION TOKENS 63	TOTAL TOKENS 16582	LLM LATENCY 2.263s	PIPELINE 2.269s	TOK/S 7328.0891
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 2	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE
connected_devices scope

SNIPPET SOURCE
connected_devices_only

RETRIEVAL
skipped_connected_devices_present

CONFIGURED TOP-K
10

EFFECTIVE TOP-K
-

INJECTED CATEGORIES
2

ContactSensor

1 services · 1 values · 0 functions

VALUES

ContactSensor_Contact

raw Contact · arg - · ret BOOL · Allows reading the value of a contact sensor device

FUNCTIONS

None

Siren

CONNECTED

file:///home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_by_category_manual_20260506/retrieval_gpt41_mini_cat3/side_by_side_report... 33/56

2 services · 1 values · 1 functions

VALUES

Siren_SirenMode

raw SirenMode · arg - · ret ENUM · Allows for the control of the Siren

FUNCTIONS

Siren_SetSirenMode

raw SetSirenMode · arg ENUM · ret VOID · Allows for the control of the Siren

GT JOICODE

cron="" · period=0

```
if ((#Entrance #ContactSensor).contactsensor_contact == true) {  
  
    (#Siren).siren_setsirenmode("emergency")  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#ContactSensor).contactsensor_contact == true) {  
    (#Siren).siren_setsirenmode("emergency")  
}
```

FAILURE REASONS

- dataflow
- gt_mismatch

LINE DIFF

GT	Generated
if ((#Entrance #ContactSensor).contactsensor_contact == true) {	if ((#ContactSensor).contactsensor_contact == true) {
	(#Siren).siren_setsirenmode("emergency")
(#Siren).siren_setsirenmode("emergency")	}
}	

Row 82 · Category 3

If it is raining, set all dehumidifiers in the house to drying mode.

지금 비가 오고 있으면, 집안의 모든 제습기를 건조 모드로 설정해줘.

GT name: IfItIsRainingSetAllDehumidifiersInHouse				
Service list: service_list_ver2.0.1.json				
Output name: SetDehumidifierToDryingWhenRaining				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.8284	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 16583
COMPLETION TOKENS 71	TOTAL TOKENS 16654	LLM LATENCY 2.112s	PIPELINE 2.119s	TOK/S 7885.7900
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
---	--	--	------------------------	----------------------	--------------------------

RainSensor

1 services · 1 values · 0 functions

VALUES

RainSensor_Rain

raw Rain · arg - · ret BOOL · A Device that senses rain

FUNCTIONS

None

Dehumidifier

2 services · 1 values · 1 functions

VALUES

Dehumidifier_DehumidifierMode

raw DehumidifierMode · arg - · ret ENUM · Allows for the control of the dehumidifier mode

FUNCTIONS

Dehumidifier_SetDehumidifierMode

raw SetDehumidifierMode · arg ENUM · ret VOID · Allows for the control of the dehumidifier mode

GT JOICODE

cron="" · period=0

```
if ((#RainSensor).rainsensor_rain == true) {  
  
    all(#House #Dehumidifier).dehumidifier_setdehumidifiermode("drying")  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#RainSensor).rainsensor_rain == true) {  
    (#Dehumidifier).dehumidifier_setdehumidifiermode("drying")  
}
```

FAILURE REASONS

dataflow

extraneous

gt_mismatch

LINE DIFF

GT	Generated
if ((#RainSensor).rainsensor_rain == true) {	if ((#RainSensor).rainsensor_rain == true) {
	(#Dehumidifier).dehumidifier_setdehumidifiermode("drying")
all(#House #Dehumidifier).dehumidifier_setdehumidifiermode("drying")	}
}	

Row 83 · Category 3

If the livingroom brightness is 200 lux or higher, turn off the livingroom light.

거실 조도가 200 럭스 이상이면, 거실 불을 꺼줘.

GT name: IfLivingroomBrightnessIs200LuxOrHigherTurn				
Service list: service_list_ver2.0.1.json				
Output name: TurnOffLivingroomLightIfBright				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9032	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 18199
COMPLETION TOKENS 64	TOTAL TOKENS 18263	LLM LATENCY 4.245s	PIPELINE 4.253s	TOK/S 4302.3393
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 2	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 3 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 3
---	--	--	------------------------	----------------------	--------------------------

LightSensor

1 services · 1 values · 0 functions

VALUES

LightSensor_Brightness
raw Brightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity

FUNCTIONS

None

Light

15 services · 8 values · 7 functions

VALUES

Light_ColorMode

raw ColorMode · arg - · ret ENUM · A numerical representation of the brightness intensity

Light_CurrentBrightness

raw CurrentBrightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentColorTemperature

raw CurrentColorTemperature · arg - · ret INTEGER · A numerical representation of the brightness intensity

Light_CurrentHue

raw CurrentHue · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentRGB

raw CurrentRGB · arg - · ret STRING · A numerical representation of the brightness intensity

Light_CurrentSaturation

raw CurrentSaturation · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentX

raw CurrentX · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentY

raw CurrentY · arg - · ret DOUBLE · A numerical representation of the brightness intensity

FUNCTIONS

Light_MoveToBrightness

raw MoveToBrightness · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToColorTemperature

raw MoveToColorTemperature · arg INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHue

raw MoveToHue · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHueAndSaturation

raw MoveToHueAndSaturation · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToRGB

raw MoveToRGB · arg INTEGER | INTEGER | INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToSaturation

raw MoveToSaturation · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToXY

raw MoveToXY · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Switch

CONNECTED

4 services · 1 values · 3 functions

VALUES

Switch_Switch

raw Switch · arg - · ret BOOL · Allows for the control of a Switch device

FUNCTIONS

Switch_Off

raw Off · arg - · ret VOID · Allows for the control of a Switch device

Switch_On

raw On · arg - · ret VOID · Allows for the control of a Switch device

Switch_Toggle

raw Toggle · arg - · ret BOOL · Allows for the control of a Switch device

GT JOICODE

cron="" · period=0

```
if ((#LivingRoom #LightSensor).lightsensor_brightness >= 200) {  
  
    (#LivingRoom #Light).switch_off()  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#LightSensor #Livingroom).lightsensor_brightness >= 200) {  
    (#Light #Livingroom).switch_off()  
}
```

FAILURE REASONS

- dataflow
- gt_mismatch
- precondition

LINE DIFF

GT	Generated
if ((#LivingRoom #LightSensor).lightsensor_brightness >= 200) {	if ((#LightSensor #Livingroom).lightsensor_brightness >= 200) {
	(#Light #Livingroom).switch_off()
(#LivingRoom #Light).switch_off()	}
}	

Row 84 · Category 3

If the temperature is 36.5 degrees or higher, set the target temperature to 30.

구역1의 온도가 36.5도 이상이면, 에어컨을 30도로 설정해줘.

GT name: IfTemperaturesIs365DegreesOrHigherSet				
Service list: service_list_ver2.0.1.json				
Output name: SetTargetTempIfHigh				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9306	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 1.0000	NUMERIC 0.3750	ENUM 1.0000	PROMPT TOKENS 16723
COMPLETION TOKENS 75	TOTAL TOKENS 16798	LLM LATENCY 2.205s	PIPELINE 2.212s	TOK/S 7618.4861
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 2	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
---	--	--	------------------------	----------------------	--------------------------

TemperatureSensor

1 services · 1 values · 0 functions

VALUES

TemperatureSensor_Temperature
raw Temperature · arg - · ret DOUBLE · Get the temperature from a Device that reports current temperature

FUNCTIONS

None

AirConditioner

4 services · 2 values · 2 functions

VALUES

AirConditioner_AirConditionerMode

raw AirConditionerMode · arg - · ret ENUM · Controls air conditioner mode and temperature settings

AirConditioner_TargetTemperature

raw TargetTemperature · arg - · ret DOUBLE · Controls air conditioner mode and temperature settings

FUNCTIONS

AirConditioner_SetAirConditionerMode

raw SetAirConditionerMode · arg ENUM · ret VOID · Controls air conditioner mode and temperature settings

AirConditioner_SetTargetTemperature

raw SetTargetTemperature · arg DOUBLE · ret VOID · Controls air conditioner mode and temperature settings

GT JOICODE

cron="" · period=0

```
if ((#Sector1 #TemperatureSensor).temperaturesensor_temperature >= 36.5) {  
  
    (#Sector1 #AirConditioner).airconditioner_settargettemperature(30)  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#TemperatureSensor).temperaturesensor_temperature >= 36.5) {  
    targetTemp = 30.0  
    (#AirConditioner).airconditioner_settargettemperature(targetTemp)  
}
```

FAILURE REASONS

- arg_type
- gt_mismatch
- numeric_grounding

LINE DIFF

GT	Generated
if ((#Sector1 #TemperatureSensor).temperaturesensor_temperature >= 36.5) {	if ((#TemperatureSensor).temperaturesensor_temperature >= 36.5) {
	targetTemp = 30.0
(#Sector1 #AirConditioner).airconditioner_settargettemperature(30)	(#AirConditioner).airconditioner_settargettemperature(targetTemp)
	}
}	

Row 85 · Category 3

If the carbon dioxide concentration in the farm is 800ppm or higher, sound the emergency siren.

농장의 이산화탄소 농도가 800ppm 이상이면, 긴급 사이렌을 울려줘.

GT name: IfCarbonDioxideConcentrationInFarmsIs800ppm				
Service list: service_list_ver2.0.1.json				
Output name: EmergencySirenOnHighCO2				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9870	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 17183
COMPLETION TOKENS 71	TOTAL TOKENS 17254	LLM LATENCY 2.292s	PIPELINE 2.299s	TOK/S 7529.2372
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 2	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
---	--	--	------------------------	----------------------	--------------------------

AirQualitySensor

7 services · 7 values · 0 functions

VALUES

AirQualitySensor_CarbonDioxide

raw CarbonDioxide · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_DustLevel

raw DustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_FineDustLevel

raw FineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

CONNECTED

AirQualitySensor_Humidity

raw Humidity · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_Temperature

raw Temperature · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_TvocLevel

raw TvocLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_VeryFineDustLevel

raw VeryFineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

FUNCTIONS

None

Siren

2 services · 1 values · 1 functions

CONNECTED

VALUES

Siren_SirenMode

raw SirenMode · arg - · ret ENUM · Allows for the control of the Siren

FUNCTIONS

Siren_SetSirenMode

raw SetSirenMode · arg ENUM · ret VOID · Allows for the control of the Siren

GT JOICODE

cron="" · period=0

```
if ((#Farm #AirQualitySensor).airqualitysensor_carbondioxide >= 800) {  
  
    (#Siren).siren_setsirenmode("emergency")  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#AirQualitySensor).airqualitysensor_carbondioxide >= 800) {  
    (#Siren).siren_setsirenmode("emergency")  
}
```

FAILURE REASONS

- dataflow
- gt_mismatch

LINE DIFF

GT	Generated
if ((#Farm #AirQualitySensor).airqualitysensor_carbondioxide >= 800) {	if ((#AirQualitySensor).airqualitysensor_carbondioxide >= 800) {
	(#Siren).siren_setsirenmode("emergency")
(#Siren).siren_setsirenmode("emergency")	}
}	

Row 86 · Category 3

If the temperature on the airquality sensor is below 20 degrees, close the door.

회의실 온도가 20도 미만이면, 회의실 문을 닫아줘.

GT name: IfTemperatureOnAirQualitySensorIsBelow20				
Service list: service_list_ver2.0.1.json				
Output name: CloseDoorIfAirQualitySensorTempBelow20				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9350	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 17216
COMPLETION TOKENS 62	TOTAL TOKENS 17278	LLM LATENCY 4.138s	PIPELINE 4.145s	TOK/S 4175.7498
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 2	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
---	--	--	------------------------	----------------------	--------------------------

AirQualitySensor

7 services · 7 values · 0 functions

VALUES

AirQualitySensor_CarbonDioxide

raw CarbonDioxide · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_DustLevel

raw DustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_FineDustLevel

raw FineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

CONNECTED

AirQualitySensor_Humidity

raw Humidity · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_Temperature

raw Temperature · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_TvocLevel

raw TvocLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_VeryFineDustLevel

raw VeryFineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

FUNCTIONS

None

Door

CONNECTED

3 services · 1 values · 2 functions

VALUES

Door_DoorState

raw DoorState · arg - · ret ENUM · Allow for the control of a door

FUNCTIONS

Door_Close

raw Close · arg - · ret VOID · Allow for the control of a door

Door_Open

raw Open · arg - · ret VOID · Allow for the control of a door

GT JOICODE

cron="" · period=0

```
if ((#MeetingRoom #AirQualitySensor).airqualitysensor_temperature < 20) {  
  
    (#MeetingRoom #Door).door_close()  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#AirQualitySensor).airqualitysensor_temperature < 20) {  
    (#Door).door_close()  
}
```

FAILURE REASONS

- dataflow
- gt_mismatch

LINE DIFF

GT	Generated
if ((#MeetingRoom #AirQualitySensor).airqualitysensor_temperature < 20) {	if ((#AirQualitySensor).airqualitysensor_temperature < 20) {
	(#Door).door_close()
(#MeetingRoom #Door).door_close()	}
}	

Row 87 · Category 3

If the light in the meeting room is off, turn it on.

회의실 조명이 꺼져 있으면, 조명을 켜줘.

GT name: IfLightInMeetingRoomIsOffTurn				
Service list: service_list_ver2.0.1.json				
Output name: TurnOnMeetingRoomLightIfOff				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9222	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 18108
COMPLETION TOKENS 60	TOTAL TOKENS 18168	LLM LATENCY 1.691s	PIPELINE 1.700s	TOK/S 10745.2094
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 2	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT					
Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.					
CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2

Light	CONNECTED
15 services · 8 values · 7 functions	
VALUES	
Light_ColorMode raw ColorMode · arg - · ret ENUM · A numerical representation of the brightness intensity	
Light_CurrentBrightness raw CurrentBrightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity	
Light_CurrentColorTemperature raw CurrentColorTemperature · arg - · ret INTEGER · A numerical representation of the brightness intensity	

Light_CurrentHue

raw CurrentHue · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentRGB

raw CurrentRGB · arg - · ret STRING · A numerical representation of the brightness intensity

Light_CurrentSaturation

raw CurrentSaturation · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentX

raw CurrentX · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentY

raw CurrentY · arg - · ret DOUBLE · A numerical representation of the brightness intensity

FUNCTIONS

Light_MoveToBrightness

raw MoveToBrightness · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToColorTemperature

raw MoveToColorTemperature · arg INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHue

raw MoveToHue · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHueAndSaturation

raw MoveToHueAndSaturation · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToRGB

raw MoveToRGB · arg INTEGER | INTEGER | INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToSaturation

raw MoveToSaturation · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToXY

raw MoveToXY · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Switch

CONNECTED

4 services · 1 values · 3 functions

VALUES

Switch_Switch

raw Switch · arg - · ret BOOL · Allows for the control of a Switch device

FUNCTIONS

Switch_Off

raw Off · arg - · ret VOID · Allows for the control of a Switch device

Switch_On

raw On · arg - · ret VOID · Allows for the control of a Switch device

Switch_Toggle

raw Toggle · arg - · ret BOOL · Allows for the control of a Switch device

GT JOICODE

cron="" · period=0

```
if ((#MeetingRoom #Light).switch_switch == false) {  
  
    (#MeetingRoom #Light).switch_on()  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#Light #MeetingRoom).switch_switch == false) {  
    (#Light #MeetingRoom).switch_on()  
}
```

FAILURE REASONS

- dataflow
- extraneous
- gt_mismatch
- precondition
- semantic

LINE DIFF

GT	Generated
if ((#MeetingRoom #Light).switch_switch == false) {	if ((#Light #MeetingRoom).switch_switch == false) {
	(#Light #MeetingRoom).switch_on()
(#MeetingRoom #Light).switch_on()	}
}	

Row 88 · Category 3

If the temperature in the bedroom is 36.5 degrees or higher, set the bedroom air conditioner to 30 degrees.

안방 온도가 36.5도 이상이면, 에어컨을 30도로 설정해줘.

GT name: IfTemperatureInBedroomIs365Degrees				
Service list: service_list_ver2.0.1.json				
Output name: BedroomTemperatureControl				
Retrieval cats: not used (connected_devices scope)				
DET 60.7099	PASS No	GT EXACT No	SIMILARITY 0.7965	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 0.0000	NUMERIC 0.5000	ENUM 1.0000	PROMPT TOKENS 16728
COMPLETION TOKENS 70	TOTAL TOKENS 16798	LLM LATENCY 1.452s	PIPELINE 1.460s	TOK/S 11567.2772
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 2	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 2 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 2
---	--	--	------------------------	----------------------	--------------------------

TemperatureSensor

1 services · 1 values · 0 functions

VALUES

TemperatureSensor_Temperature
raw Temperature · arg - · ret DOUBLE · Get the temperature from a Device that reports current temperature

FUNCTIONS

None

AirConditioner

4 services · 2 values · 2 functions

VALUES

AirConditioner_AirConditionerMode

raw AirConditionerMode · arg - · ret ENUM · Controls air conditioner mode and temperature settings

AirConditioner_TargetTemperature

raw TargetTemperature · arg - · ret DOUBLE · Controls air conditioner mode and temperature settings

FUNCTIONS

AirConditioner_SetAirConditionerMode

raw SetAirConditionerMode · arg ENUM · ret VOID · Controls air conditioner mode and temperature settings

AirConditioner_SetTargetTemperature

raw SetTargetTemperature · arg DOUBLE · ret VOID · Controls air conditioner mode and temperature settings

GT JOICODE

cron="" · period=0

```
if ((#Bedroom #TemperatureSensor).temperaturesensor_temperature >= 36.5) {  
  
    (#Bedroom #AirConditioner).airconditioner_settargettemperature(30)  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#TemperatureSensor #bedroom).temperaturemeasurement_temperature >= 36.5) {  
    (#AirConditioner #bedroom).airconditioner_settargettemperature(30.0)  
}
```

FAILURE REASONS

- dataflow
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- numeric_grounding
- service_match
- unknown_service:temperaturemeasurement_temperature

LINE DIFF

GT	Generated
if ((#Bedroom #TemperatureSensor).temperaturesensor_temperature >= 36.5) {	if ((#TemperatureSensor #bedroom).temperaturemeasurement_temperature >= 36.5) {
	(#AirConditioner #bedroom).airconditioner_settargettemperature(30.0)
(#Bedroom #AirConditioner).airconditioner_settargettemperature(30)	}
}	

Row 89 · Category 3

If motion is being detected in the living room, turn on the living room light.

거실에 움직임이 감지되고 있으면, 조명을 켜줘.

GT name: IfMotionIsBeingDetectedInLivingRoomTurn				
Service list: service_list_ver2.0.1.json				
Output name: LivingRoomMotionLightOn				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9060	GT SERVICE COV. 1.0000
RECEIVER COV. 1.0000	DATAFLOW 0.2000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 18194
COMPLETION TOKENS 60	TOTAL TOKENS 18254	LLM LATENCY 2.057s	PIPELINE 2.065s	TOK/S 8874.5199
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 2	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 3 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 3
---	--	--	------------------------	----------------------	--------------------------

MotionSensor

1 services · 1 values · 0 functions

VALUES

MotionSensor_Motion

raw Motion · arg - · ret BOOL · Motion sensor device

FUNCTIONS

None

Light

15 services · 8 values · 7 functions

VALUES

Light_ColorMode

raw ColorMode · arg - · ret ENUM · A numerical representation of the brightness intensity

Light_CurrentBrightness

raw CurrentBrightness · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentColorTemperature

raw CurrentColorTemperature · arg - · ret INTEGER · A numerical representation of the brightness intensity

Light_CurrentHue

raw CurrentHue · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentRGB

raw CurrentRGB · arg - · ret STRING · A numerical representation of the brightness intensity

Light_CurrentSaturation

raw CurrentSaturation · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentX

raw CurrentX · arg - · ret DOUBLE · A numerical representation of the brightness intensity

Light_CurrentY

raw CurrentY · arg - · ret DOUBLE · A numerical representation of the brightness intensity

FUNCTIONS

Light_MoveToBrightness

raw MoveToBrightness · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToColorTemperature

raw MoveToColorTemperature · arg INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHue

raw MoveToHue · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToHueAndSaturation

raw MoveToHueAndSaturation · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToRGB

raw MoveToRGB · arg INTEGER | INTEGER | INTEGER · ret VOID · A numerical representation of the brightness intensity

Light_MoveToSaturation

raw MoveToSaturation · arg DOUBLE · ret VOID · A numerical representation of the brightness intensity

Light_MoveToXY

raw MoveToXY · arg DOUBLE | DOUBLE · ret VOID · A numerical representation of the brightness intensity

Switch

CONNECTED

4 services · 1 values · 3 functions

VALUES

Switch_Switch

raw Switch · arg - · ret BOOL · Allows for the control of a Switch device

FUNCTIONS

Switch_Off

raw Off · arg - · ret VOID · Allows for the control of a Switch device

Switch_On

raw On · arg - · ret VOID · Allows for the control of a Switch device

Switch_Toggle

raw Toggle · arg - · ret BOOL · Allows for the control of a Switch device

GT JOICODE

cron="" · period=0

```
if ((#LivingRoom #MotionSensor).motionsensor_motion == true) {  
  
    (#LivingRoom #Light).switch_on()  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#MotionSensor #LivingRoom).motionsensor_motion == true) {  
    (#Light #LivingRoom).switch_on()  
}
```

FAILURE REASONS

- dataflow
- gt_mismatch
- semantic

LINE DIFF

GT	Generated
if ((#LivingRoom #MotionSensor).motionsensor_motion == true) {	if ((#MotionSensor #LivingRoom).motionsensor_motion == true) {
	(#Light #LivingRoom).switch_on()
(#LivingRoom #Light).switch_on()	}
}	

Row 90 · Category 3

If the lab humidity is 50 or higher, turn on the dehumidifier, and if it's below 50, turn on the humidifier.

연구실 습도가 50 이상이면 제습기를 켜고 50 미만이면 가습기를 켜줘.

GT name: IfLabHumidityIs50OrHigherTurnOn				
Service list: service_list_ver2.0.1.json				
Output name: HumidityControlLab				
Retrieval cats: not used (connected_devices scope)				
DET 68.6366	PASS No	GT EXACT No	SIMILARITY 0.6773	GT SERVICE COV. 0.5000
RECEIVER COV. 0.5000	DATAFLOW 0.2000	NUMERIC 0.7500	ENUM 1.0000	PROMPT TOKENS 17206
COMPLETION TOKENS 104	TOTAL TOKENS 17310	LLM LATENCY 4.806s	PIPELINE 4.814s	TOK/S 3601.5230
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 4 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 4
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HumiditySensor

1 services · 1 values · 0 functions

VALUES

HumiditySensor_Humidity
raw Humidity · arg - · ret DOUBLE · Allow reading the relative humidity from devices that support it

FUNCTIONS

None

Dehumidifier

2 services · 1 values · 1 functions

VALUES

Dehumidifier_DehumidifierMode

raw DehumidifierMode · arg - · ret ENUM · Allows for the control of the dehumidifier mode

FUNCTIONS

Dehumidifier_SetDehumidifierMode

raw SetDehumidifierMode · arg ENUM · ret VOID · Allows for the control of the dehumidifier mode

Switch

CONNECTED

4 services · 1 values · 3 functions

VALUES

Switch_Switch

raw Switch · arg - · ret BOOL · Allows for the control of a Switch device

FUNCTIONS

Switch_Off

raw Off · arg - · ret VOID · Allows for the control of a Switch device

Switch_On

raw On · arg - · ret VOID · Allows for the control of a Switch device

Switch_Toggle

raw Toggle · arg - · ret BOOL · Allows for the control of a Switch device

Humidifier

CONNECTED

2 services · 1 values · 1 functions

VALUES

Humidifier_HumidifierMode

raw HumidifierMode · arg - · ret ENUM · Maintains and sets the state of an humidifier

FUNCTIONS

Humidifier_SetHumidifierMode

raw SetHumidifierMode · arg ENUM · ret VOID · Maintains and sets the state of an humidifier

GT JOICODE

cron="" · period=0

```
if ((#Lab #HumiditySensor).humiditysensor_humidity >= 50) {  
  
    (#Lab #Dehumidifier).switch_on()  
  
} else {  
  
    (#Lab #Humidifier).switch_on()  
  
}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#HumiditySensor).humiditysensor_humidity >= 50) {  
    (#Dehumidifier).dehumidifier_setdehumidifiermode("dehumidifying")  
} else if ((#HumiditySensor).humiditysensor_humidity < 50) {  
    (#Humidifier).humidifier_sethumidifiermode("auto -")  
}
```

FAILURE REASONS

- dataflow
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- numeric_grounding

LINE DIFF

GT	Generated
if ((#Lab #HumiditySensor).humiditysensor_humidity >= 50) {	if ((#HumiditySensor).humiditysensor_humidity >= 50) {
	(#Dehumidifier).dehumidifier_setdehumidifiermode("dehumidifying")
(#Lab #Dehumidifier).switch_on()	} else if ((#HumiditySensor).humiditysensor_humidity < 50) {
	(#Humidifier).humidifier_sethumidifiermode("auto -")
} else {	}
(#Lab #Humidifier).switch_on()	
}	

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